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The art of joining: Challenging planar joints in robotically printed ceramic assemblies

M. Smigielska
ETH Zurich, Zürich, Switzerland

S. Pain
The Royal Danish Academy, Copenhagen, Denmark

M. Rafikova & J. Tobar Martinez
ETH Zurich, Zürich, Switzerland

ABSTRACT: This paper investigates novel interlocking joint systems for 3D-printed ceramic architectural elements, examining their design, fabrication, and tectonics from technical and cultural perspectives. The study tackles challenges in precision and structural stability arising from clay's non-uniform deformation during the entire fabrication process, including printing and drying. To mitigate these issues, the project relies on the concept of digital craftsmanship, manifested through iterative design and prototyping, mixed fabrication model (digital and manual) that sensibly splits the tasks between man and machine based on the knowledge and required task and a deformation-aware design to production workflow which incorporates a comprehensive set of analogue and digital methods to improve the precision of the ceramic elements by minimizing material deformation. Structural tests evaluated adhesive and dry assemblies, resulting in finger and hinge joints designed for efficient, non-load-bearing interlocking bricks. Culturally, this research reinterprets joinery as both functional and expressive, challenging traditional brick and masonry aesthetics. The project results with a 1x2m perforated ceramic wall demonstrator, made of 19 stoneware elements and proposes an architectural language that merges traditional craftsmanship with digital techniques, creating a sophisticated dialogue between the human, machine and the material.

1 INTRODUCTION

Masonry is one of the oldest building crafts with the earliest examples dating back to 8000BC. Throughout this time, the design of ceramic assemblies has followed technical innovation and cultural evolution. Yet, despite these developments, the basic techniques of joining brick, stone, concrete or ceramic materials have remained unchanged. In wet construction, elements are bonded together with cement or mortar into a permanent, fused, composite body, and in dry construction, elements are stacked and/or mechanically attached to a substructure in the form of pins, clips, screws or bolts.

The tectonics of different types of joints and resulting expressions are often intertwined with innovation in building techniques and a particular period's cultural and artistic movements. The refinement of additive manufacturing and the ongoing digital revolution present architects and engineers with new challenges on both technical and formal fronts.

With rapid development of 3d printing technologies, clay extrusion gained more interest in architectural research and practice. By providing geometric freedom and precision in the deposition of material, it has the potential to generate new design languages which cannot be

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achieved alone by the human hand or through traditional casting and extrusion practices. While new typologies have been explored in brick design, the joinery between the 3d printed bricks remain underexplored and maintains a traditional character, dominated by linear joints and an abundance of mortar when wet construction is employed. 3D printed ceramic elements will most likely remain discrete and be limited in size due to kiln volume limitations and material rheology causing the heavy material to collapse during the printing process. Therefore, tectonic innovation in ceramic structures is as important as the brick itself. This calls for more research into structures with minimised use of mortar and novel joinery systems.

To this end, this project explores the design and assembly possibilities in a deformation-aware and materially optimised, lightweight, self-supporting ceramic brick wall with a novel joinery system. The primary goal of the investigation was to examine the non-planar joint to improve stability, ease of assembly and novel aesthetics it offers while in tandem exploring strategies to combat shrinkage and deformation of the material during processing. This paper describes this process through the lens of craft, that focuses on the considerations around the articulation and materiality of the joints. Testing of different geometries are described as well as the use of mortar, which is used as a bonding material that binds the history of traditional masonry to the newfound forms of 3D printed ceramic elements.

2 STATE OF THE ART

2.1 *Joinery techniques in 3D printed ceramics*

Joining 3D-printed ceramic pieces in architectural assemblies is challenging due to the precision required in finish and alignment coupled with the deformation that occurs in the processing of clay, driving researchers to explore novel solutions.

The research directions mostly include traditional methods to account for imprecision of the parts- either joining through adhesive bonding or employing a substructure while maintaining gaps between the elements. The Hive project (Shi et al. 2019) proposed a spatially complex interlocking system of hexagonal brick for a large-scale wall, however, the final assembly was secured with a considerable amount of a wet adhesive to maintain structural stability and mitigate the imprecision of the fired modules.

Other initiatives include using non-structural connectors, tested in Ceramic Information (Lange and Holohan 2019) which utilize standardized metal bolts to stack over 1000 components into double curved wall or custom 3D-printed PLA pads to absorb the friction between the ceramic-ceramic connection in the Ficus column project (Figueiredo et al. 2019).

Very few attempts were done in mono material self-supporting or load-bearing ceramic structures, which demonstrate the complexity of the tolerance problem. Reefs startup used integrated tongue-and-groove joints to stack bricks underwater into modular construction (rrreefs, 2020). In "Brick by Bit" a wall is constructed from double sided bricks interlocked in the center, which maintains stability without the use of mortar (Roznovski, 2024). Both projects assume large offsets to accommodate high tolerances. While these projects explore self-supporting systems, the contact surfaces between the units remain planar and do not benefit from the geometric freedom offered by 3d printing technology. In the project Textile Column, an interlocking non-planar tectonic system was developed for the vertical connection of ceramic tiles as a fire protection around a circular structural steel column (Pain et al. 2022). While this project failed to deal with issues of accumulative tolerance, it opens an interesting research avenue for non-planar joints that require a high level of tolerance at an architectural scale. Furthermore, the state of the art indicates more research is needed to exploit the geometric freedom of 3d printed ceramics and how it can be applied in self-supporting and load bearing structures, where the compression strength of ceramic units is taken into account.

2.2 *Strategies for imprecision in 3D printed ceramic fabrication: Design for tolerances and material control*

All clays are susceptible to shrinkage due to moisture loss during the drying and firing process. Common approaches to addressing this issue involve scaling up the piece in proportion to the expected shrinkage percentage during firing. Additionally, post-processing of fired pieces is often used, particularly in traditional tile production, to achieve greater precision, though this method is labour-intensive (Bechthold et al., 2015).

Unfortunately, those methods do not apply easily to 3d printed modules, which present higher geometric complexity and uneven distribution of material, overall leading to non-uniform shrinkage across the piece. Therefore, effective joining of complex modules requires careful planning from the design stage, accounting for material and geometric transformations inherent to the fabrication process.

The Ceramic Information Pavilion team refined brick designs through iterative prototyping. To address geometric inaccuracies, they used an offset double-line design for structural stability, while bolt holes with added tolerance provided flexibility during assembly. (Lange and Holohan 2019). The initial printing tests of "Ficus" revealed significant misalignment of the parts due to differential shrinkage between the top and bottom sections of the column. The authors propose a computational model that analyses the geometry and makes the necessary modifications, ensuring the final model accurately represents the physical version of the digital design. (Figueiredo et al., 2019). In the "Hive" project, digital scans revealed deviations between the model and printed units caused by clay's plasticity and slanted masonry angles. Global deformation and slumping ruled out zero-tolerance approaches. Instead, interlocking units printed upside-down were used ensuring compatibility despite the material's unpredictable behaviour.

Using AI for material behaviour prediction has recently gained interest for deformed metal sheets (Zwierzycki et al. 2017), metal rods (Smigielska 2018) or adaptive robotic wood carving (Brugnaro et al. 2017). AI coupled with clay was briefly approached at UCL (Chellani 2023) with cGANs or with Neural Networks at IAAC (Hanlon 2023) to predict the shrinkage of bricks with identical outer but varying infill. However, the details provided are insufficient to assess the scalability and applicability of these methods to other brick typologies. Another approach includes a physics-based finite element simulation tool of the time dependent moisture level in a ceramic 3D printed part (Subedi & Suresh 2024). While it may help avoiding failures and cracking during drying, at this moment the tool cannot predict the deformation itself.

In summary, these projects employed various methods to improve, but not fully solve the material deformation challenges. The variety and fragmentation of those attempts highlight the need to further search through different approaches. In this project we focus on building a comprehensive set of rules around clay behaviour control through all design to fabrication phases, as well as on developing joinery systems for higher tolerances.

3 METHODS

The project employed the concepts of digital craftsmanship manifested on three layers: (1) *design through research* exhibited through iterative design and prototyping, (2) *mixed fabrication model* (digital and manual) that sensibly splits the tasks between man and machine based on the knowledge and required task and (3) *deformation-aware design to production workflow*, which incorporates analogue and digital methods to improve the precision of the ceramic elements by minimizing material deformation.

3.1 *Digital design and fabrication infrastructure*

The project utilized well established digital design-to-fabrication workflow in the Rhino Grasshopper environment with multiple parameters controlling the design and fabrication

settings, like the tolerance offsets, the slicing layer height, print path resolution and its direction, extruder and robot speed, to name a few. Furthermore, it provides printability feedback, by graphical representation of overhang levels and checking the required material amount and bricks size to match the clay cartridge and printing area.

The fabrication setup consisted of a 6 axis ABB 1200 robot arm and customized WASP XL clay extruder with 4mm nozzle. A 7-liter clay cartridge was suspended on a custom-built gantry with a balancer to offload this weight from the robot with a limited payload. A stoneware clay body with 25% chamotte 0–0.2mm was used, diluted with 40 ml water/kg and 4 drops of dispex/ kg to create a printable material consistency. The printed modules were placed under plastic sheeting until bone dry, and then fired in an electric kiln to a temperature of 1250 °C.

3.2 *Digital craftsmanship*

3.2.1 *Mixed fabrication model*

The overall production process was divided into the following phases: (1) material preparation (2) file generation and 3d printing, (3) drying the pieces, (4) firing, and (5) assembly. The process required human assistance at each step and human intervention significantly improved the precision and production success rates due to the volatility of the process. For example, the material in the cartridges tended to dry with time and water accumulated at the bottom of the cartridge due to gravity. This led to uneven material flows during printing, which could be compensated with adapting extrusion speed "on the go" by an operator upon visual inspection. Depending on the root of the problem, one can adapt the extruder speed for minor extrusion adaptations, robot speed or pressure level in the clay cartridge for a more significant intervention. Printed pieces dry on porous racks and under perforated plastic sheeting for better air circulation, but they are also supervised by a craftsperson and adjusted to get more or less airflow depending on the speed of the drying. The digital fabrication process of 3D printing itself takes only 30 minutes per piece, which is a fraction of time in the whole process from printing, to drying to firing of 7 days in total.

3.2.2 *Deformation- aware design to fabrication workflow*

While the precision of 3D printed ceramic elements remains a major obstacle in the development of more sophisticated assemblies, the project proposes a holistic approach to minimise material deformation during the whole design-to-production process with digital and analogue interventions, as follows: (1)interlocking joint systems which create flexible spaces that maintain cohesion, allowing slight deformations without compromising overall stability, (2) minimising joint areas, (3) the use of curved geometries as a rule across the design of the entire element, (4) a maximum 50 degree geometric inclination to prevent excessive overhangs, (5) limiting the brick size to match the amount of clay of one cartridge, (6) optimized material mixture to minimize the amount of water with a deflocculant added to reduce shrinkage, (7) a grogged clay that improves stability during printing, (8) every second printing path had a reversed order, which prevents the prints from being 'dragged' with one repetitive path direction and therefore improves the stability and precision during printing, and (9) the printing speed was adjusted on the go to maintain the uniform extrusion rate across the piece.

The Drying process incorporated several interventions to improve the elements precision; (10) intermittent flipping the pieces 180 degrees after 24 hours as they tend to dry faster on the upper extremities, (11) the use of plastic cover to slow down the evaporation, and (12) the use of proximity drying to aid uniform deformation in relation to neighbouring elements.

3.3 *Designing through research- Joint studies*

The experiments are presented as 3 consecutive studies of partial ceramic assemblies with a focus on the geometry of the joints and bricks as well as on the drying strategy.

3.3.1 *Study 1*

In the first study 3 types of geometries were tested: Type 1 – an element with an undulating planar joint and types 2 and 3, kinematically constrained V-shaped and freeform joints respectively (Figure 1). The elements were flipped 90º from the print plane to easily obtain non-planar joining surfaces. To obtain better precision of the elements, we used a proximity drying technique where pieces are dried in place, so they shrink and deform in correspondence with the neighbouring element. The result of this experiment demonstrated that Type 2, the V-shaped joint, appeared to show less deformation when assembled than the other 2 types. The V-shaped connections created 2 smaller planes on the connecting surfaces which broke up the larger, flat plane. It was observed through types 1 and 3 that deformation is more obvious across a larger planar surface, and when that surface is broken down into smaller faces, deformation patterns are interrupted.

While the results showed that deformation could be controlled to some degree through proximity drying and the V shaped joint, it was apparent that a purely dry stacked system would not be feasible. Since the elements were relatively lightweight, a wall of them would not withstand lateral forces. In addition, tiny point loads occurred throughout the assemblies as a result of the elements' rotation from the prints original orientation meaning that each had an undulating surface native to the expression of printed clay. To develop the system into a practical joint, mortar was introduced as a bonding material. Mortar was chosen for its historical and pragmatic relevance, in order to fix and stabilise the assembly and mediate both accumulated and local tolerances. The pieces were fired again to a higher temperature of 1250º C to increase their strength and a series of tests were run to evaluate the feasibility of a mortared joint. The structural tests were conducted on a small assembly of 6 bricks ca 45x25x15cm cm size (WxHxD), organized by 2 pieces on 3 vertical layers, with uniformly applied top load. The results of these studies concluded that the assemblies were able to withstand a force of up to 150kg without rupture or cracking.

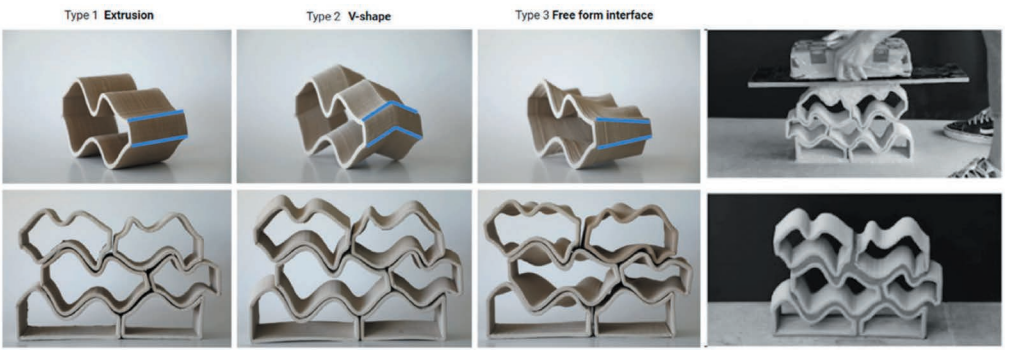


Figure 1. Left: Studies of 3 types of joints and analysis of deformation post firing. Right: structural tests of bricks with mortar.

3.3.2 *Study 2*

Following the success of the V shape connection and the use of mortar, the second study focused on how this connection could be further articulated by minimising contact surface ratio to create smaller planes for deformation. This was done in tandem with considering the aesthetics of a filigree bonded wall system as a departure from a solid masonry wall.

A further set of 3 geometries were developed from the V shaped connection and were analysed across several parameters including amount of material required, contact surface ratio, assembly logic and aesthetic value (Figure 2). Of the 3, the third iteration was selected to study as a series of printed elements due to a minimal surface connection, conservative use of material as well it being the most radical departure from the planar surfaces of the traditional brick.

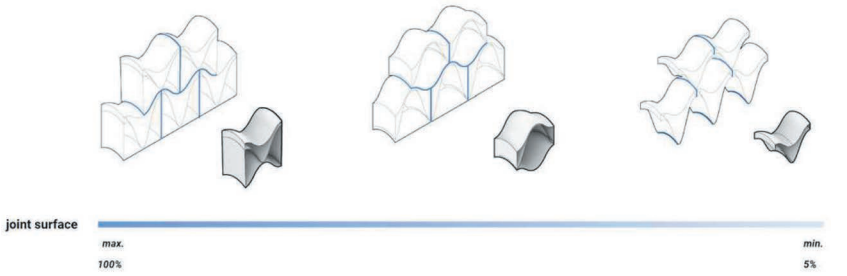


Figure 2. Digital studies looking at minimizing joint surface between the elements.

Even though the previous tests showed that breaking down larger surfaces broke the deformation pattern, the contact surface was developed into a curve as opposed to a peak. This compromise was taken as the aesthetics of the wall and the element was favourable to continuous curvature and we wanted to test whether this adjustment would still perform.

As it was not possible to dry this series vertically, due to small contact surfaces and the fragility of the elements as was done in the previous experiment, these elements were dried horizontally. The drying process was carried under the plastic cover and flipped over once leather hard.

In terms of shrinkage and deformation, it was found that minimal deformation occurred over the curved connection which resulted in a good connection between the upper and lower elements in the series. The planar, linear connections at the upper parts of each element were less successful and were difficult to accurately position in the assembly. This was less to do with deformation issues and more to do with the fact that none of the joints were effectively 'hidden' as was possible with the non-planar connection. In terms of aesthetic, and relationship to masonry, this assembly presented both aspects of the traditional linear bond (in the upper joints) as well as the potential of a more integrated and aggregated expression (in the lower joint).

3.3.3 *Study 3*

The findings from the second study determined a third and final set of iterations with greater focus on the upper two joints of each piece. It was the intention that these joints might follow the same logic of a curved geometry; favourable to shrinkage and deformation, an interlocking system; to give a greater sense of synthesis across the aesthetics of the assembly, as well as provide a better construction logic that would support the application of mortar. A further set of geometries were developed capitalising on a curved interface across the upper joints. From these iterations, a final geometry was developed resulting in a more complex finger-like joint where the elements overlapped each other leaving space for the application of mortar in-between. The curved surfaces required a rotational interlocking movement in the construction phase due to the elimination of all planar surfaces (Figure 3).

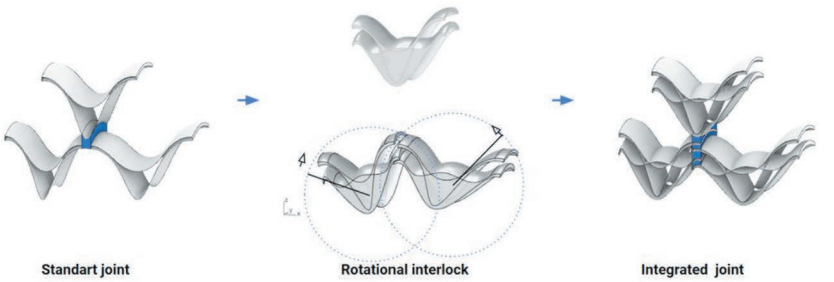


Figure 3. The integrated joint required rotational interlock due to the elimination of all planar surfaces.

The final iteration was selected for testing on one hand for its compelling aesthetic of a more integrated assembly, and on the other, following the previous findings that meant we could use the method of proximity drying. A custom design tool was developed to analyse and optimise the overall expression of the final joint design on a larger scale for a self-supporting, non-load-bearing wall. This tool allows for adjusting the overall size of the wall and its vertical and horizontal density based on input graph within ranges which are optimal for successful printability and deformation. This graph defined the height and width of each brick based on the parametric distribution of two intersecting sine curves. This allows for various local porosity levels ranging from 20% to 80%, depending on how much and where the visual privacy is needed.

3.4 *Processing and assembly*

After printing and drying the pieces to a leather-hard state, they were arranged on a horizontal board in their final configuration, with overlapping extremities placed near neighbouring pieces. This assembly allowed the large lattice-like structure to dry and deform as a cohesive body rather than as smaller, more vulnerable individual elements.

The final pieces were mortared together in sequence, layer by layer. The connections meant the element could be interlocked by rotating them on the Y axis to fit with the corresponding piece. Like the articulation of traditional masonry, there were a multitude of ways in which the mortar could be expressed. The decision was taken to set the mortar line back from the outer boundary of each element enhancing the filigree lattice like structure of the wall. The colour of the mortar was comparable to the colour of the ceramic pieces which resulted in an expression less like the elemental quality of a brick wall and more akin to a homogenous structure (Figure 4).

4 RESULTS AND EVALUATION

The result is a 1m x 1.6m lattice-like structure composed of 19 bespoke elements that presents non-planar, interlocking finger and hinge joints bonded with mortar. Fabrication and assembly were completed over the course of one week, from material preparation to final assembly. The geometric form of the elements was derived from observations on how curved surfaces mitigate deformation in robotically printed ceramic elements, creating a unique aesthetic where joints and bricks merge into a unified geometric canvas.

The structure faced two main challenges: deformation during firing, which hindered precise interlocking of the elements, and issues in the construction phase where additional elements were added before the mortar was dry leading to the layers slumping. The firing issue could have been addressed using spacers, small pre-fired elements that control deformation during firing, and the construction issue could have been mitigated using formwork, such as timber or metal pins, to support the structure during assembly. By introducing these techniques, deformation and slumping could have been mitigated. From a material efficiency perspective, the demonstrator successfully met its optimisation goals. The careful use of mortar at the minimised joint surfaces allowed for a reduced adhesive quantity, thereby improving overall material efficiency.

The project presents a novel tectonic and aesthetic qualities in 3D printed assembled ceramics. The structurally sound joinery system and targeted design and fabrication methods allowed us to achieve sophisticated design results while meeting precision requirements with additional benefits of material use optimisation. The formal language of the prototype and its articulation provide an example of how the aesthetics and understanding of traditional ornamental brickwork and lattice structures can be reinterpreted using the geometric freedom of printing using robots. While the project does not fully solve the inherent problem of clay deformation during ceramic processing, it presents effective methods to minimise this issue and integrate the material's behaviour into joint design for robotically printed assemblies.

The project highlights the role of the craftsperson in the digital fabrication processes. 3D printing with clay is an example of a process heavily dependent on human experience, intuition and tacit knowledge of the material and its processing. As such, 3D-printed ceramics represent a continuation of traditional craft, further enhanced with digital technologies rather than an industrialised, automated approach focused on minimising human intervention. The process represents a well-balanced human-robot collaboration for clay 3d printing that deals collectively with fabrication errors towards a common goal of bringing the physical object as close as possible to their digital twin. While the robot brings the high geometric precision on the level of extrusion of each unique brick, the craftsperson performs very well to account for their low precision on the material level. At large, this strategy facilitates the integration of volatile materials into semi-automated architectural production.



Figure 4. Construction details and the homogenous material expression of the final piece.

5 CONTRIBUTION

The project advances the topic of tectonics, material precision and digital craftsmanship in assembly of 3d printed ceramic elements. The main contributions are: (1) a novel joinery system (finger and hinge joint) for higher tolerances, new tectonics, aesthetics and optimised use of materials (both clay and mortar), (2) deformation-aware design-to-fabrication workflow crafted for ceramic robotic 3D printing as a comprehensive digital and analogue set of methods for a better clay deformation control.

6 OUTLOOK

The outlook of this project falls into two main areas: ceramic joints and their structural performance, and computational methods for material behaviour prediction. This research has the potential to enrich the design space of ceramic joints, and consequently significantly open up the expression of larger architectural elements. To fully implement the system, further advancements are necessary: evaluating the structural performance of ceramic-ceramic and ceramic-mortar interfaces under loads exceeding 150 kg and integrating custom structural analysis into digital design tools. This integration could enable performance-based navigation for new joint and design systems, addressing the demand for high-tolerance 3D-printed tectonics. Additionally, better control and prediction of clay deformation is crucial for expanding design possibilities. A predictive model integrated into the design workflow could accelerate iterations through simulations, reducing the need for physical prototyping. This could be achieved using advanced computational strategies like machine learning, which encodes material behaviour and human intuition.

REMARKS

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